

Divyanshu Kumar

Python Backend & AI Engineer

work.kumardivyanshu@gmail.com — +91 7970479325

LinkedIn — GitHub — Portfolio

PROFESSIONAL SUMMARY

Python Backend & AI Engineer with 1+ year of experience building scalable backend services, automation pipelines, and AI-driven applications. Experienced in REST API development, structured workflow automation, RAG-based GenAI systems using LangChain, and real-time system integration. Strong understanding of service architecture, modular design, and cloud-ready deployments.

EDUCATION

B.Tech in Computer Science Engineering

Dumka Engineering College, 2025

CGPA: 8.55/10

EXPERIENCE

Tech Genius Institute

Aug 2025 – Present

AI & Software Engineer / Mentor

- Designed Python backend services integrating real-time sensor inputs, control logic, and AI inference pipelines for robotics applications.
- Built REST APIs and structured automation scripts enabling reliable communication between hardware modules and software systems.
- Implemented modular project architecture with logging, debugging, and testing workflows to improve maintainability.
- Led end-to-end execution from requirement analysis to deployment-ready prototypes following Agile practices.

Gyantu (Remote)

Nov 2024 – Jan 2025

AI Intern

- Developed modular Python backend for conversational AI using LLaMA 3.2, LangChain, and ChromaDB with service-oriented structure.
- Implemented Retrieval-Augmented Generation (RAG) pipelines including document ingestion, embedding creation, and vector search optimization.
- Designed agent-style workflows using LangChain chains and tool integrations for dynamic query routing.
- Built Streamlit interface with clear backend–frontend separation, improving usability and response accuracy by 40%.

TechInterex (Remote)

Feb 2025 – Apr 2025

Data Science Intern

- Built structured Python data pipelines for preprocessing, feature engineering, and experimentation across ML projects.
- Automated model evaluation, result tracking, and reporting dashboards for efficient iteration cycles.
- Collaborated on predictive modeling and recommendation system implementations.

PROJECTS

Conversational AI Chatbot (GenAI + RAG)

- Built Python backend service enabling document-aware Q&A using LangChain, ChromaDB, and LLaMA 3.2.
- Designed automated ingestion pipeline for document parsing, embedding generation, and semantic retrieval.
- Implemented agent-style chains for context-aware query refinement and response orchestration.

AI-Powered Driver Monitoring System

- Developed real-time inference backend using Flask integrating CNN and 3D-CNN models for drowsiness and behavior detection.
- Integrated IoT modules (NodeMCU, ultrasonic, IR sensors) using MQTT-based communication for real-time vehicle feedback.
- Designed end-to-end pipeline from sensor input to decision logic and system alerts.

Gene Ontology Classification

- Engineered preprocessing pipeline for high-dimensional biological datasets with feature filtering and encoding.
- Implemented TabTransformer-based deep learning models achieving improved classification performance.
- Explored Positive-Unlabeled learning strategies for handling severe class imbalance.

TECHNICAL SKILLS

Languages: Python, SQL, C++

Backend: Flask, REST APIs, Service Architecture, Modular Design

Automation & Data: Selenium, BeautifulSoup, Data Pipelines, ETL Concepts

GenAI: LangChain, RAG, LLaMA, Transformers, Agentic Workflows

DevOps: Git, Linux, Docker (Basics), CI/CD Concepts, AWS/GCP Fundamentals

Messaging: MQTT, Background Processing Concepts

ACHIEVEMENTS

Winner — Innovathon (BIT Sindri): Smart Grocery Quality Control System

Finalist — HackFest '24 (IIT ISM): Unified Customer Insights

Best Final Year Project — AI Driver Assistance System

Solved 250+ Data Structures & Algorithms Problems